随机模拟方法第十次作业

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1. 求证: 自由随机游走在连续极限下的概率密度 $W(x,t)$ 满足

$$\begin{cases} \dfrac{\partial W}{\partial t} = D\dfrac{\partial^2 W}{\partial x^2}, & x \in \mathbb{R}, t > 0, \\ W(x,0) = \delta(x), & x \in \mathbb{R}. \end{cases}$$

解答: 利用

$$W(x,t) = \frac{1}{\sqrt{4\pi D t}} \mathrm{e}^{-\frac{x^2}{4Dt}}$$

直接计算可得

$$\frac{\partial W}{\partial t} = D\frac{\partial^2 W}{\partial x^2}, \quad \forall\, x \in \mathbb{R}, t > 0.$$

而 $W(\cdot,t)$ 为 $\mathcal{N}(0,2Dt)$ 的密度函数, 并且 $W(\cdot,t)\,\mathrm{d}x \Rightarrow \delta_0, t \to 0$, 故 $W(x,0) = \delta(x), \forall\, x \in \mathbb{R}$. □

2. 设 $x_1 > 0$, 求证: 具有反射边界的随机游走在连续极限下的概率密度 $W_r(x,t)$ 满足

$$\begin{cases} \dfrac{\partial W_r}{\partial t} = D\dfrac{\partial^2 W_r}{\partial x^2}, & x \leqslant x_1, t > 0, \\ W_r(x,0) = \delta(x), & x \leqslant x_1, \\ \dfrac{\partial W_r}{\partial x}(x_1,t) = 0, & t > 0. \end{cases}$$

解答: 注意到 $\forall\, x \leqslant x_1, t > 0$, 有 $W_r(x,t) = W(x,t) + W(2x_1 - x, t)$, 结合第一题可得

$$\frac{\partial W_r}{\partial t} = D\frac{\partial^2 W_r}{\partial x^2}, \quad \forall\, x \leqslant x_1, t > 0,$$

并且 $W_r(x,0) = \delta(x), \forall\, x \leqslant x_1$. 同时

$$\frac{\partial W_r}{\partial x}(x_1,t) = \frac{\partial W}{\partial x}(x_1,t) - \frac{\partial W}{\partial x}(x_1,t) = 0, \quad \forall\, t > 0.$$

□

3. 设 $x_1 > 0$, 求证: 具有吸收边界的随机游走在连续极限下的概率密度 $W_a(x,t)$ 满足

$$\begin{cases} \dfrac{\partial W_a}{\partial t} = D\dfrac{\partial^2 W_a}{\partial x^2}, & x \leqslant x_1, t > 0, \\ W_a(x,0) = \delta(x), & x \leqslant x_1, \\ W_a(x_1,t) = 0, & t > 0. \end{cases}$$

解答: 注意到 $\forall\, x \leqslant x_1, t > 0$, 有 $W_a(x,t) = W(x,t) - W(2x_1 - x, t)$, 结合第一题可得

$$\frac{\partial W_a}{\partial t} = D\frac{\partial^2 W_a}{\partial x^2}, \quad \forall\, x \leqslant x_1, t > 0,$$

并且 $W_a(x,0) = \delta(x), \forall\, x \leqslant x_1$. 同时

$$W_a(x_1,t) = W(x_1,t) - W(x_1,t) = 0, \quad \forall\, t > 0.$$

□

4. 写出反正弦定律证明中数学归纳法的详细步骤.

解答: 我们已经得到了

$$u_{2k} = \sum_{r=1}^{k} f_{2r} u_{2(k-r)}, \quad \forall\, k \in \mathbb{Z}^+,$$

$$P_{0,2n} = P_{2n,2n} = u_{2n}, \quad P_{2k,2n} = \frac{1}{2}\sum_{r=1}^{k} f_{2r} P_{2(k-r),2(n-r)} + \frac{1}{2}\sum_{r=1}^{n-k} f_{2r} P_{2k,2(n-r)}, \quad \forall\, 0 < k < n.$$

下面对 n 归纳证明 $\forall\, 0 \leqslant k \leqslant n$, 有 $P_{2k,2n} = u_{2k}u_{2(n-k)}$. 当 $n = 1$ 时, 由于 $u_0 = 1$, 故结论已得证. 假设结论对小于 n 的正整数均成立, 对 $n \geqslant 2, \forall\, 0 < k < n$, 有

$$\begin{aligned} P_{2k,2n} &= \frac{1}{2}\sum_{r=1}^{k} f_{2r} P_{2(k-r),2(n-r)} + \frac{1}{2}\sum_{r=1}^{n-k} f_{2r} P_{2k,2(n-r)} \\ &= \frac{1}{2}\sum_{r=1}^{k} f_{2r} u_{2(k-r)} u_{2(n-k)} + \frac{1}{2}\sum_{r=1}^{n-k} f_{2r} u_{2k} u_{2(n-k-r)} \\ &= \frac{1}{2} u_{2(n-k)} \sum_{r=1}^{k} f_{2r} u_{2(k-r)} + \frac{1}{2} u_{2k} \sum_{r=1}^{n-k} f_{2r} u_{2(n-k-r)} \\ &= \frac{1}{2} u_{2(n-k)} u_{2k} + \frac{1}{2} u_{2k} u_{2(n-k)} = u_{2k} u_{2(n-k)}. \end{aligned}$$

故结论对 n 也成立, 由数学归纳法知结论成立. □